



DGN_KISTAR

Modified settings for KISTAR

2026.06.15
Giwon Baek

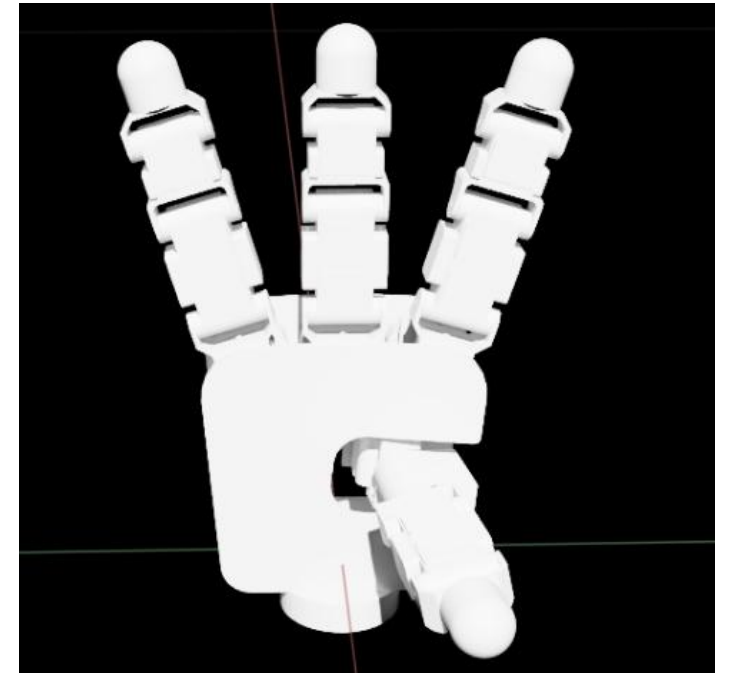
- Object scale normalization & augmentation
- Initial pose
- Contact points
- Self penetration keypoints
- Weights
- Validation

■ Object scale normalization & augmentation

- Objects: ContactDB (50), YCB (20) — scale normalization required
- Normalize all objects to fit within a unit sphere (diameter: 1 m).
- Apply random scaling augmentation with target sizes of 6, 9, 12, and 15 cm.

■ Initial pose

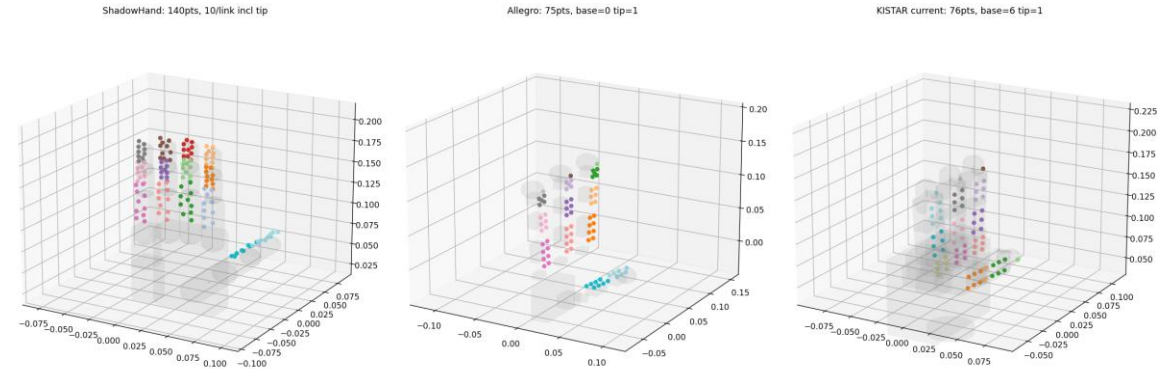
- Initial pose
 - Thumb: $\text{joint}_0 = +81, \text{joint}_1 = -72$
 - Index: $\text{joint}_0 = -12$
 - Ring: $\text{joint}_0 = +12$
- To achieve force closure, the thumb moves toward the middle finger, while the index and ring fingers perform abduction.
- Since DGN adds jitter to the initial joint positions, the joints are not set exactly at their limits.



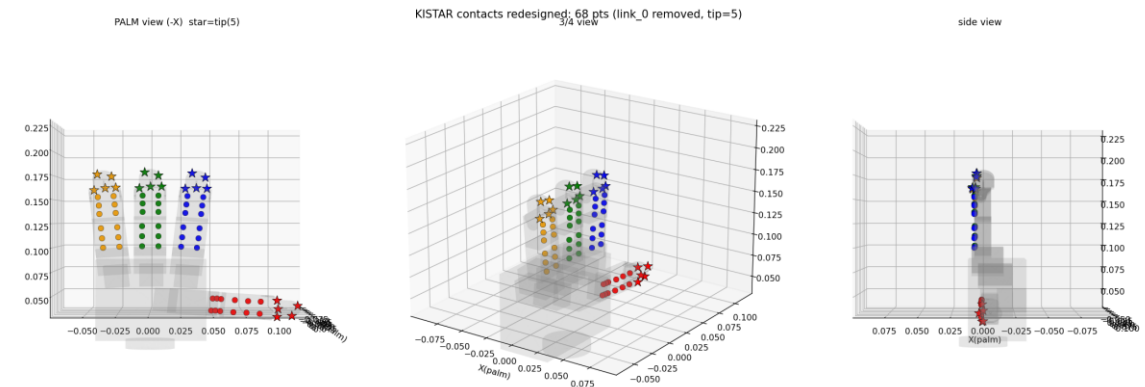
Initial pose
visualization

■ Contact points

- Limited to the palm-facing surface.
- Similar to Allegro, link_0 is not defined.
- For all other links, 6 points are assigned per link, and 5 points are assigned to each fingertip.
- Total: 68 points.



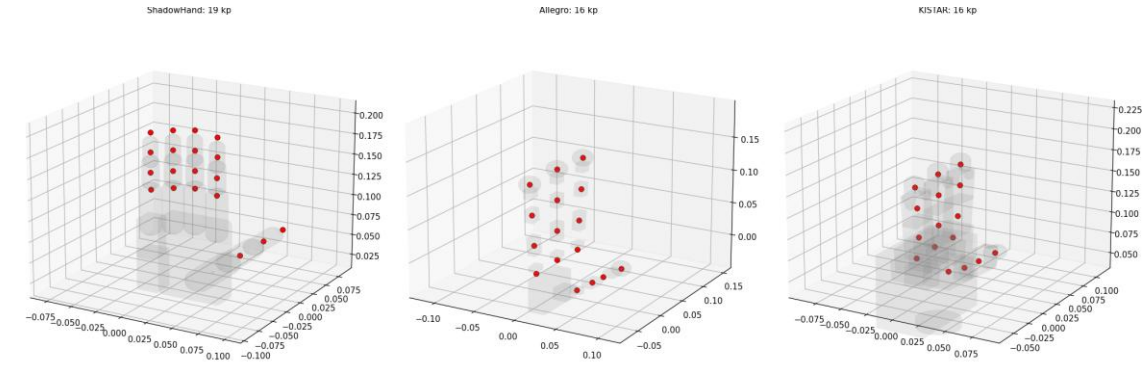
Shadow, Allegro and former KISTAR contact point visualization



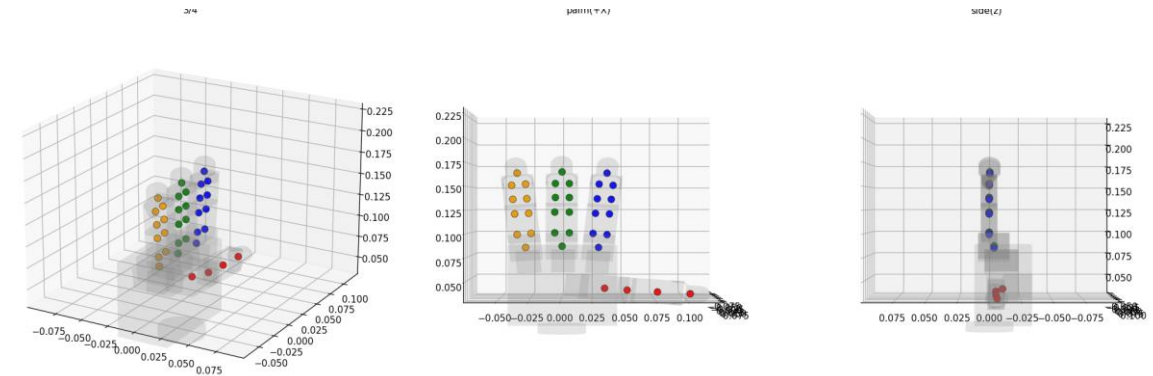
Final KISTAR contact point visualization

■ Self penetration keypoints

- Keypoints are placed more densely on each link, instead of using a single point at the link center.
- Self-penetration is masked out between keypoints belonging to the same link or adjacent links.
- Only one keypoint is assigned to link_0 and the thumb, as they are less prone to self-penetration.



Shadow, Allegro and former KISTAR self penetration point visualization



Final KISTAR self penetration point visualization

■ Weights

- Increase the weights assigned to the joint-limit and self-penetration penalties.

Weights	Base	DGN_KISTAR
w_fc	1	1
w_dis	100	100
w_pen	100	100
w_joints	1	20
w_spen	10	20

■ Validation

- The previous approach performed both penetration checking and Isaac Gym-based collision checking.
- In DGN_KISTAR, joint-limit violations and self-penetration are evaluated too.
- Although the paper defines a valid grasp as having a maximum penetration below 0.1 cm, the implementation actually uses the criterion $E_{\text{pen}} < 0.001$.
- To relax this condition, we adopt $E_{\text{pen}} < 0.01$ as the decision threshold.

Filter
$E_{\text{pen}} < 0.01$
$E_{\text{joints}} < 0.001$
$E_{\text{spen}} < 0.005$

Thank you

Center for Humanoid Research
26.06.15
Giwon Baek
Contact: 01gwbaek@naver.com